Iteratively reweighted least squares

We want to solve $\underbrace{\sum_{i=1}^{n} x_{ij}\{y_i - F(\underline{x}_i\underline{\beta})\} = 0}_{\text{score equation}}$ for $j = 0, \ldots, p-1$

Take a first order Taylor approximation of F about $\underline{x}_i\underline{\beta}^*$

$$F(\underline{x}_i\underline{\beta}) \approx F(\underline{x}_i\underline{\beta}^*) + F'(\underline{x}_i\underline{\beta}^*)(\underline{x}_i\underline{\beta} - \underline{x}_i\underline{\beta}^*)$$

$$= F(\underline{x}_i\underline{\beta}^*) + F(\underline{x}_i\underline{\beta}^*)(1 - F(\underline{x}_i\underline{\beta}^*))\underline{x}_i(\underline{\beta} - \underline{\beta}^*)$$

Noting $F(\underline{x}_i\underline{\beta}^*) = p_i^*$, then

$$F(\underline{x}_i\underline{\beta}) \approx p_i^* + p_i^*(1-p_i^*)\underline{x}_i(\underline{\beta} - \underline{\beta}^*)$$

Substituting backs into the score equation

$$\sum_{i=1}^{n} x_{ij}\{y_i - p_i^* - p_i^*(1-p_i^*)\underline{x}_i(\underline{\beta} - \underline{\beta}^*)\} = 0$$

$$\sum_{i=1}^{n} x_{ij} \underbrace{p_i^*(1-p_i^*)}_{\text{weights } w_i} \left\{ \underline{x}_i\underline{\beta}^* + \underbrace{\frac{y_i - p_i^*}{p_i^*(1-p_i^*)}}_{z_i - \text{working response}} - \underline{x}_i\underline{\beta} \right\} = 0$$

$$\sum_{i=1}^{n} x_{ij} w_i (z_i - \underline{x}_i\underline{\beta}) = 0 \quad \leftarrow \text{system of linear equations.}$$

In matrix form

$$\underline{\underline{X}}^T \underline{\underline{W}}^* \underline{\underline{X}}\,\underline{\beta} = \underline{\underline{X}}^T \underline{\underline{W}}^* \underline{z}^*$$

$\underline{\underline{W}}^*$: diagonal matrix with $w_{ii}^* = p_i^*(1-p_i^*)$

This equation gives the solution to the weighted least squares problem $\operatorname{argmin} \sum w_{ii}(z_i - \underline{x}_i\underline{\beta})^2$

Prospective study – units are sampled and the covariates measured, outcomes then observed going forward.

Retrospective study – we sample cases based on the outcomes and then record the covariates. In this case sampling proportions depend on the outcome.

For the retrospective study, we introduce dummy variables $Z_i \in \{0, 1\}$ defined such that $Z_i = 1$ if unit i is included in the study. The sampling proportions are

$$\pi_1 = \mathbb{P}(Z_i = 1 \mid Y_i = 1) \quad \text{and} \quad \pi_0 = \mathbb{P}(Z_i = 1 \mid Y_i = 0)$$

The sampling proportions depend on Y but $\underline{x}$. Using Bayes rule

$$\mathbb{P}(Y_i = 1 \mid Z_i = 1) = \frac{\mathbb{P}(Z_i = 1 \mid Y_i = 1)\mathbb{P}(Y_i = 1)}{\mathbb{P}(Z_i = 1 \mid Y_i = 1)\mathbb{P}(Y_i = 1) + \mathbb{P}(Z_i = 1 \mid Y_i = 0)\mathbb{P}(Y_i = 0)}$$

only interested in those in the study

$$= \frac{\pi_1 \cdot F(\underline{x}_i\beta)}{\pi_1 F(\underline{x}_i\beta) + \pi_0 (1 - F(\underline{x}_i\beta))}$$

If $\underline{x}_i\beta = \beta_0 + \beta_1 x_{i,1} + \cdots$ then $\beta_0^* = \beta_0 + \log(\pi_1/\pi_0)$

$$= \frac{\pi_1 F(\underline{x}_i\beta)/(1 - F(\underline{x}_i\beta))}{\pi_1 F(\underline{x}_i\beta)/(1 - F(\underline{x}_i\beta)) + \pi_0}$$

$$= \frac{\pi_1 \exp(\underline{x}_i\beta)}{\pi_1 \exp(\underline{x}_i\beta) + \pi_0}$$

$$= \frac{(\pi_1/\pi_0)\exp(\underline{x}_i\beta)}{(\pi_1/\pi_0)\exp(\underline{x}_i\beta) + 1}$$

$$= \frac{\exp(\underline{x}_i\beta + \log(\pi_1/\pi_0))}{1 + \exp(\underline{x}_i\beta + \log(\pi_1/\pi_0))}$$